

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                     | Mark  | Comments |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------|
| 05.1     | 1. To kill any fungus/bacteria on surface of seeds or in soil;<br>2. So only the added fungus has any effect;                                                                                                                                                                                                                        | 2     |          |
| 05.2     | So that only nitrate or ammonia/type of fertiliser affects growth;                                                                                                                                                                                                                                                                   | 1     |          |
| 05.3     | 1. So that effects of nitrate or ammonium alone could be seen;<br>2. So that effects of fungus can be seen;                                                                                                                                                                                                                          | 2     |          |
| 05.4     | 1. Weigh samples at intervals during drying;<br>2. To see if weighings became constant (by 3 days);                                                                                                                                                                                                                                  | 2     |          |
| 05.5     | With live fungus – showing effects of the fungus:<br>1. Fungus increases growth of roots and shoots in both;<br>2. Produces greater growth with nitrate;<br>With heat-treated fungus – showing effects of fertiliser:<br>3. Similar dry masses for roots and shoots;<br>4. (Probably) no significant difference because SDs overlap; | 4     |          |
| 05.6     | 1. Dry mass measures/determines increase in biological/organic material;<br>2. Water content varies;                                                                                                                                                                                                                                 | 2     |          |
| 05.7     | 1. Fungus with nitrate-containing fertiliser gave largest shoot: root ratio;<br>2. And largest dry mass of shoot;<br>3. 6.09:1 compared with ammonium-containing fertiliser 4.18:1;                                                                                                                                                  | 2 max |          |